Multivariate Probability Distributions

Univariate distributions $\rightarrow$ deal with a Single RV

Multivariate " $\rightarrow$ deal with two or more RVs simultaneously.

↳ Allow for the study of Correlation and dependence between variables.

$$X \sim N(\mu_X, \sigma_X^2) \rightarrow X \text{ is a univariate Gaussian.}$$

Prob. Density Function :

$$\begin{cases} p_X(x) = \dfrac{1}{\sqrt{2\pi\sigma_X^2}} \cdot e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \\ \\ p_Y(y) = \dfrac{1}{\sqrt{2\pi\sigma_Y^2}} \cdot e^{-\frac{(y-\sigma_Y)^2}{2\sigma_Y^2}} \end{cases}$$

$$Y \sim N(\mu_Y, \sigma_Y^2) :$$

For two independent events A and B, | $P(A)$ $P(B)$

$$\begin{cases} P(A \cap B) = P(A \text{ "and" } B) \\ \qquad\qquad = P(A) \cdot P(B) \end{cases}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A) \Rightarrow P(A \cap B) = P(B) \cdot P(A)$$

↳ If A and B are independent,

Occurrence of B doesn't affect A.

$$p_{XY}(x,y) = p_X(x) \cdot p_Y(y) = \text{Joint PDF}$$

$$= \frac{1}{\sqrt{(2\pi)^2 \sigma_X^2 \sigma_Y^2}} \cdot e^{-\left[\frac{(x-\mu_X)^2}{2\sigma_X^2} + \frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right]}$$

$$= \frac{1}{(2\pi)^{2/2} \cdot \underbrace{\left(\sigma_X^2 \sigma_Y^2\right)^{1/2}}_{|\Sigma|^{1/2}}} \cdot e^{-\frac{1}{2} \underbrace{\left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]}_{(x-\mu)^T \Sigma^{-1} (x-\mu)}}$$

$$\Sigma = \text{Covariance Matrix}$$

$$= \begin{bmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{bmatrix}$$

$$\Sigma^{-1} = \begin{bmatrix} 1/\sigma_X^2 & 0 \\ 0 & 1/\sigma_Y^2 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \overbrace{\begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} x_1 \\ a_{22} x_2 \end{bmatrix} = a_{11} x_1^2 + a_{22} x_2^2$$

$$(x-y)^T A (x-y) = a_{11} (x_1 - y_1)^2 + a_{22} \underline{(x_2 - y_2)}^2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \qquad A = \Sigma^{-1} = \begin{pmatrix} 1/\sigma_x^2 & 0 \\ 0 & 1/\sigma_y^2 \end{pmatrix}$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$$

$$(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \frac{(x - \mu_x)^2}{\sigma_x^2} + \frac{(y - \mu_y)^2}{\sigma_y^2}$$

$$\Sigma = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix}, \quad \det(\Sigma) = |\Sigma| = \sigma_x^2 \sigma_y^2$$

$$\boxed{p_{xy}(x, y) = \frac{1}{2\pi \, |\Sigma|^{1/2}} \cdot e^{-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})}}$$

↳ PDF of a Bivariate Gaussian Distribution

Variance of a random variable:

$$\sigma_X^2 = E\{(X-\mu_X)^2\}$$

$$\sigma_Y^2 = E\{(Y-\mu_Y)^2\}$$

$$Cov(X,Y) = E\{(X-\mu_X)(Y-\mu_Y)\}$$

↳ measures the degree to which the RVs X and Y change together.

- Positive Covariance: when $X\uparrow \Rightarrow Y\uparrow$
- Negative Covariance: when $X\uparrow \Rightarrow Y\downarrow$
- Zero Covariance: No linear relationship between X and Y.

Vectored RV $\Rightarrow$ $\underbrace{\text{A vector of univariate RVs}}$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$(x-\mu)(x-\mu)^T = \begin{bmatrix} (x_1-\mu_1) \\ (x_2-\mu_2) \end{bmatrix} \begin{bmatrix} (x_1-\mu_1) & (x_2-\mu_2) \end{bmatrix}$$

2×1 (1×2)

$$= \begin{bmatrix} (x_1-\mu_1)^2 & (x_1-\mu_1)(x_2-\mu_2) \\ (x_2-\mu_2)(x_1-\mu_1) & (x_2-\mu_2)^2 \end{bmatrix}$$

$$\text{Covariane Matrix} = E\{(x-\mu)(x-\mu)^T\}$$

$$= \begin{bmatrix} E\{(x_1-\mu_1)^2\} & E\{(x_1-\mu_1)(x_2-\mu_2)\} \\ E\{(x_2-\mu_2)(x_1-\mu_1)\} & E\{(x_2-\mu_2)^2\} \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_{x_1}^2 & \underline{Cov(x_1, x_2)} \\ \underline{Cov(x_2, x_1)} & \sigma_{x_2}^2 \end{bmatrix}$$

$Cov(x_1, x_2) = Cov(x_2, x_1)$

$$\Sigma = \begin{bmatrix} \sigma_{x_1}^2 & cov(x_1, x_2) \\ Cov(x_1, x_2) & \sigma_{x_2}^2 \end{bmatrix} \rightarrow \text{Always Symmetric}$$

↳ Is a positive semidefinite matrix
i.e. the eigenvalues are nonnegative.

Can you prove this?

$$\Sigma = \begin{bmatrix} \sigma_{x_1}^2 & 0 \\ 0 & \sigma_{x_2}^2 \end{bmatrix}$$

$Cov(X_1, X_2) = 0$

i.e. X_1 and X_2 are <u>INDEPENDENT</u>